

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer with production MLOps pipeline experience - FastAPI/Docker packaging, evaluation audit trails, and LLM grounding in healthcare - seeking to help W&B customers accelerate adoption of ML experiment tracking and LLM evaluation at scale. Master's degree (Kent State University). Track record: ~35% retrieval precision gain in production healthcare RAG, >30% hallucination reduction in agentic workflows, 15-20% recall gains on high-risk patient categories, and ~30% fewer post-deployment defects. Skilled in Python, PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Docker-based deployment, and LLMOps instrumentation. Experienced building and communicating ML systems to both technical practitioners and non-technical stakeholders.

CORE COMPETENCIES

MLOps / LLMOps • ML Evaluation Pipelines • PyTorch • Hugging Face Transformers • scikit-learn / XGBoost • RAG Systems • Agentic LLM Workflows • FastAPI / Docker • Observability & Logging • Grafana • CI/CD (Jenkins) • Technical Onboarding & Documentation

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **Retrieval-Augmented Generation (RAG)** systems for clinical knowledge retrieval; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~**35%** and reduced irrelevant retrieval by >**30%**. Retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, grounding rules, and evaluation guardrails; improved agent response stability by ~**25%**. Maintained comprehensive **evaluation audit trails** for LLMOps governance.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by **15-20%** on high-risk categories via class weighting, stratified sampling, and threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~**30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts; raised dataset reliability above **98%**. Maintained HIPAA-conscious data governance and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP | Hyderabad, India

- Optimized SQL and **T-SQL stored procedures** powering enterprise Order-to-Cash platform; reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >**30%** and improved release cycle time ~**35-40%**.
- Designed performance dashboards using **SQL Server DMVs and Grafana** for long-running queries and I/O contention; cut incident recurrence ~**25%** and improved root-cause resolution speed ~**30%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics | Hyderabad, India

- Built a **transformer-based video summarization** system across 5,000+ recorded sessions; reduced manual review time by **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%** and raised contextual answer accuracy >**85%**.
- Deployed as a **Flask-based API** with lightweight interface for non-technical staff; authored technical onboarding materials for staff adoption.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with **audit-ready reasoning traces** - mirrors W&B's traceability and evaluation-first philosophy.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI system using Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose for pose conditioning, and an **LLM-based agent orchestrator**. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker. Experiment tracking, hyperparameter sweeps, and model checkpointing managed across training iterations.

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a unified **YOLOv8** model; reduced inference latency **~30%**. LabelImg annotations, augmentation pipeline, sliding-window confidence aggregation, and NMS tuning - demonstrates systematic ML experimentation and evaluation discipline.

IoT + ML Smart Building Indoor Temperature Prediction

Arduino + DHT11 sensor pipeline with PLX-DAQ logging; time-series forecasting with **walk-forward validation** comparing Random Forest, SVR, and Linear Regression. Lag-window feature engineering; Random Forest achieved **~20-30% lower MAE** vs linear baseline. Structured comparative evaluation across model variants.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React

ML Libraries: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, LLMops, embeddings, vector search

Computer Vision: YOLOv8, OpenCV, ControlNet, Stable Diffusion

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps, RESTful APIs, observability/logging, cloud-ready deployment

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required